FOOD DELIVERY TIME PREDICTION

Final Project Report

Data Analysis, Clustering and Machine Learning

Project Area	Data Science / Machine Learning
Dataset	Food_Delivery_Time_Prediction.csv
Number of Records	200
Original Variables	15
Main Task	Predict Fast or Delayed delivery
Methods	K-Means, Hierarchical Clustering, Logistic Regression, Neural Network

1. Introduction

Food delivery looks simple from the customer's side, but the actual delivery time can depend on many things. The distance between the restaurant and customer, traffic, weather, order priority, delivery-person experience and the time of the order can all have an effect. The purpose of this project is to study these factors using a real-looking food delivery dataset and then use machine learning to understand delivery patterns and predict whether an order is likely to be fast or delayed.

The project follows a step-by-step approach. First, the data is checked and prepared. Next, useful features are created from the location and order-time information. Clustering is then used to find groups of similar orders. Finally, Logistic Regression and a simple Feedforward Neural Network are used for the Fast/Delayed classification task.

2. Objectives

- Understand the structure of the food delivery dataset.
- Check and prepare the data before applying machine learning.
- Calculate geographic distance from customer and restaurant coordinates.
- Create a rush-hour feature from the order-time information.
- Use K-Means and Hierarchical Clustering to find groups of similar deliveries.
- Predict whether a delivery is Fast or Delayed.
- Compare a simple Logistic Regression model with a Feedforward Neural Network.

- Study how traffic, weather, distance, priority and delivery-person experience relate to delivery time.

3. Dataset Description

The dataset contains 200 food orders and 15 original columns. It includes information about the order, the customer and restaurant locations, delivery conditions, ratings and the final delivery time.

Column	Description
Order_ID	Unique order identifier
Customer_Location	Latitude and longitude of the customer
Restaurant_Location	Latitude and longitude of the restaurant
Distance	Given distance for the order
Weather_Conditions	Weather during the order
Traffic_Conditions	Traffic condition
Delivery_Person_Experience	Experience level of delivery person
Order_Priority	Priority of the order
Order_Time	Time category of the order
Vehicle_Type	Vehicle used for delivery
Restaurant_Rating	Restaurant rating
Customer_Rating	Customer rating
Delivery_Time	Actual delivery time
Order_Cost	Cost of the order
Tip_Amount	Tip amount

4. Data Preprocessing

The first step was to inspect the dataset. The supplied data contains 200 rows and 15 columns. No missing values or duplicate records were found in the dataset. The preprocessing part of the notebook still includes missing-value handling so that the analysis can work safely if a missing value is present.

The location columns contain latitude and longitude values in text form. These values were separated into individual latitude and longitude columns. Categorical variables such as weather, traffic, order priority, order time and vehicle type are converted into numerical columns using one-hot encoding before the prediction models are trained.

For the classification models, the data is divided into training and testing parts using an 80:20 split. The scaling values are calculated only from the training data and then applied to the test data. This avoids using information from the test set while preparing the model.

5. Feature Engineering

Two additional features were created. The first is the Haversine distance, which calculates the straight-line geographic distance between the restaurant and customer coordinates. The second is a Rush_Hour feature. In the notebook, morning and evening orders are marked as rush-hour orders.

The delivery-time target is also converted into two classes. Orders with delivery time at or below the dataset median are labelled Fast (0), while orders above the median are labelled Delayed (1). The median delivery time in the supplied dataset is about 72.78 minutes.

6. Exploratory Analysis

The average delivery time in the dataset is about 70.50 minutes. The delivery times range from approximately 15.23 minutes to 119.67 minutes. The average supplied distance is about 11.50 units, with a median of about 10.27.

The notebook also compares delivery time with traffic, weather, order priority, distance and delivery-person experience. These comparisons are useful because they show that delivery time is not controlled by a single factor.

7. K-Means Clustering

K-Means clustering was used to divide the orders into groups with similar characteristics. The clustering uses distance, calculated Haversine distance, delivery-person experience, restaurant rating, customer rating, order cost, tip amount and rush-hour information.

The features are standardized before clustering so that variables with larger numerical values do not dominate the distance calculation. The Elbow Method is used to inspect different values of K. In the project notebook, four clusters are selected for the final K-Means analysis.

After clustering, the average delivery time of each cluster is calculated. This makes it possible to see whether some groups of orders have noticeably different delivery-time patterns.

8. Hierarchical Clustering

Hierarchical clustering is used as a second way of grouping the same type of order information. The method starts with individual data points and gradually joins similar groups together. A dendrogram is included in the notebook to show this merging process.

The resulting groups are compared with the K-Means groups. Using two clustering methods gives a broader view of the structure present in the delivery data instead of depending on only one grouping method.

9. Fast or Delayed Prediction

For the prediction part, the target variable is Fast or Delayed. Logistic Regression is used as a simple baseline classification model. A Feedforward Neural Network is also included using TensorFlow/Keras.

The neural network contains an input layer followed by two hidden layers with 16 and 8 neurons. ReLU activation is used in the hidden layers and sigmoid activation is used in the output layer because the task has two classes. The model is trained using the Adam optimizer and binary cross-entropy loss.

The models are evaluated using accuracy, precision, recall and F1-score. Confusion matrices are also generated so that the number of correct and incorrect Fast/Delayed predictions can be seen directly.

10. Findings from the Dataset

The supplied dataset shows some variation in average delivery time across traffic and weather categories. In the earlier analysis of the same dataset, average delivery time was approximately 71.93 minutes for Low traffic, 70.98 minutes for Medium traffic and 67.57 minutes for High traffic. This does not follow the simple expectation that higher traffic must always mean slower delivery, which shows why the dataset should be interpreted carefully.

Weather conditions also show differences in average delivery time. The earlier analysis gave approximately 73.06 minutes for Rainy, 69.79 for Cloudy, 69.19 for Snowy and 69.48 for Sunny conditions. These differences should be treated as patterns in this particular dataset rather than as a general rule about food delivery.

The project therefore suggests that the available variables do not completely describe the real delivery process. Factors such as restaurant preparation time, actual road traffic, route choice and live GPS information are not fully represented in the supplied data.

11. Results and Discussion

The main purpose of the models is not only to produce a number but also to understand how well the available data can support prediction. The notebook reports accuracy, precision, recall and F1-score for both Logistic Regression and the neural network after execution.

Because the final submission notebook is designed to be run in the user's own Python environment, the model output values are intentionally not hard-coded into this report. This keeps the report consistent with the actual run of the final notebook and avoids presenting a result that was not generated by that execution.

12. Limitations

- The dataset contains only 200 records, so the models have a relatively small amount of data to learn from.
- The supplied distance and location information may not represent the exact route taken by the delivery person.
- Real-time traffic and weather changes are not available.
- Restaurant preparation time is not included.
- The target Fast/Delayed definition is based on the median delivery time of this dataset.
- The calculated Haversine distance represents straight-line geographic distance, not road-network travel distance.

13. Suggestions for Improvement

- Collect a larger number of delivery records.
- Include restaurant preparation time.

- Use live traffic information and route-based distance.
- Include more detailed delivery-person and vehicle information.
- Add GPS-based route information where available.
- Try additional machine-learning models and compare their performance.
- Use real-time data if the system is later converted into an application.

14. Future Scope

The project can be extended into a practical delivery-time prediction system. A future version could receive the restaurant location, customer location, current traffic, weather and order details in real time. The system could then estimate the expected delivery category before the order is dispatched. A dashboard or mobile/web application could also be developed to display delivery predictions and identify orders that may need attention.

15. Conclusion

This project provides a complete introduction to analysing food delivery data with Python and machine learning. The dataset was inspected and prepared, geographic distance and rush-hour information were created, and both clustering and classification methods were applied.

K-Means and Hierarchical Clustering were used to understand groups of similar orders. Logistic Regression and a Feedforward Neural Network were used for the Fast/Delayed prediction task. The analysis also compared delivery time with traffic, weather, priority, distance and delivery-person experience.

Overall, the project shows how data science can be used to study delivery operations. At the same time, the results highlight the importance of collecting better operational data if a more accurate real-world prediction system is required.

16. References

- Food_Delivery_Time_Prediction.csv — dataset supplied for the project.
- Food Delivery Time Prediction Jupyter Notebook — project implementation.
- Pandas and NumPy — data processing and numerical calculations.
- Matplotlib and Seaborn — data visualization.
- TensorFlow/Keras — feedforward neural network implementation.

Note: This report is written in straightforward project-report language and is based on the supplied dataset and final notebook structure. Model scores should be taken from the final notebook after running all cells successfully.